

Lecture 7 & 8

- Populaⁿ params of μ & σ^2

Random sample $\rightarrow X_1, X_2, \dots, X_n \Rightarrow \bar{X}, S^2, \hat{p}$

RV have their own sampling distribuⁿ $\xrightarrow{RV}$

$\hookrightarrow \chi^2, t, F$

- Chi-Square distribuⁿ. $Y \sim \chi^2_v$

Y has χ^2 distribuⁿ with v degree of freedom

spl. case of gamma distribuⁿ density $f(y) = \frac{1}{2^{v/2} \Gamma(v/2)} y^{v/2-1} e^{-y/2}, y > 0$

$$\Gamma(\alpha) = \int_0^\infty y^{\alpha-1} e^{-y} dy.$$

- $Y \sim \chi^2_v \Rightarrow \begin{cases} E[Y] = v, \text{Var}(Y) = 2v \\ M_Y(t) = (1-2t)^{-v/2} \end{cases} \quad t < 1/2$
Moment generating funcⁿ.

if v is small, the graph is strongly right skewed concentrated at 0, & a long right tail.

if v is large, skewness decreases, distribuⁿ becomes bell-shaped. χ^2_v becomes normal.

- $Y_i \sim N(\mu, \sigma^2), Z_i = \frac{Y_i - \mu}{\sigma}, Z_i \sim N(0, 1)$

$$\sum_{i=1}^n Z_i^2 = \sum_{i=1}^n \left(\frac{Y_i - \mu}{\sigma} \right)^2 \sim \chi^2_n$$

- if $Z_1, \dots, Z_n \stackrel{iid}{\sim} N(0, 1)$
then, $Z_1^2 + \dots + Z_n^2 \sim \chi^2_n$

- if $Y_1 \sim \chi^2_{v_1}, Y_2 \sim \chi^2_{v_2}$ then $Y_1 + Y_2 \sim \chi^2_{v_1+v_2}$

- $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2), S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$

$$\text{then } \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$$

also $\bar{X}$ & S^2 is independent, when distribuⁿ is normal.

- why do we get $(n-1)$ df.

$$\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 \sim \chi_n^2$$

$$X_i - \mu = (X_i - \bar{X}) + (\bar{X} - \mu)$$

$$\sum_{i=1}^n (X_i - \mu)^2 = \sum_{i=1}^n (X_i - \bar{X})^2 + n(\bar{X} - \mu)^2$$

divide by σ^2

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0, 1)$$

$$\Rightarrow \frac{n(\bar{X} - \mu)^2}{\sigma^2} \sim \chi_1^2$$

$$\sum \frac{(X_i - \bar{X})^2}{\sigma^2} = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

Cochran's Theorem

$$\boxed{\text{df} = n-1}$$

$$- \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2 \quad E[\chi_{n-1}^2] = n-1$$

$$E\left[\frac{(n-1)S^2}{\sigma^2}\right] = n-1 \Rightarrow \frac{n-1}{\sigma^2} E[S^2] = n-1$$

$$\therefore E[S^2] = \sigma^2$$

S^2 is the unbiased estimator of population var.

- Chi-sq. is not sym. . So $\chi_{\alpha, n}^2$ refers to a value having specified tail area.

$$\text{Right tail} \rightarrow P(\chi_n^2 > c) = \alpha$$

Ex. avg. 3 years with $\sigma = 1$

5 lifetimes $\Rightarrow 1.9, 2.4, 3.0, 3.5, 4.2$

$$\bar{x} = \frac{15}{5} = 3$$

$$S^2 = \frac{n \sum x_i^2 - (\sum x_i)^2}{n(n-1)}$$

$$S^2 = 0.815$$

$$H_0: \sigma^2 = 1, \quad \chi^2 = \frac{(n-1) s^2}{\sigma^2} = \frac{4(0.815)}{1}$$

$$\chi^2 = 3.26, \quad df = 4$$

$$\chi^2_{0.025, 4} \approx 0.484$$

$$\chi^2_{0.097, 4} \approx 11.143$$

$$0.484 < 3.26 < 11.143$$

There is no evidence here that SD differs from 1g.

- if $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ then $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

$$\therefore, Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

but if σ is unknown, then replace by s . $\frac{\bar{X} - \mu}{s/\sqrt{n}}$

no this statistic is not exactly std. normal.
This is 't-distribution'.

$$- Z \sim N(0, 1), \quad V \sim \chi^2_v \quad \left[T = \frac{Z}{\sqrt{V/v}} \sim t_v \right]$$

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1) \quad \& \quad \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$$

we know $\bar{X}$ & s^2 are indep.

$$\text{then, } \left[T = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1} \right]$$

- Prop. of t-distribution

$$1. \text{ sym. } f(t) = f(-t), \quad \left[t_{\alpha, v} = -t_{1-\alpha, v} \right]$$

2. centered at 0.

3. heavier tails than normal

4. small df $\rightarrow$ heavy tails, flatter distribution
large df $\rightarrow$ $s \sim \sigma$, t becomes normal

eg. $\mu = 500$, $n = 25$, $\bar{x} = 518$, $s = 40$
 $-t_{0.05} \leq t_{0.05}$

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}} = \frac{518 - 500}{40/\sqrt{25}} = \frac{18}{8} = 2.25$$

$$df = n - 1 = 24$$

$$t_{0.05, 24} \approx 1.711$$

$$2.25 \notin (-1.711, 1.711)$$

sample is not consistent

- if σ is known $\rightarrow Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$

if σ is unknown & normal populaⁿ

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

F

- $U \sim \chi^2_{v_1}$, $V \sim \chi^2_{v_2}$ independent

$$F = \frac{U/v_1}{V/v_2} \sim F_{v_1, v_2}$$

→ takes only nonneg. values, generally right skewed, $F > 0$, not sym.

$$\begin{aligned} X_1, \dots, X_n &\sim N(\mu_1, \sigma_1^2) & (n-1)s_1^2 &\sim \chi^2_{n-1} \\ Y_1, \dots, Y_n &\sim N(\mu_2, \sigma_2^2) & (n-1)s_2^2 &\sim \chi^2_{n-1} \end{aligned}$$

$$\frac{s_1^2/\sigma_1^2}{s_2^2/\sigma_2^2} \sim F_{n-1, n-1}$$

if $\sigma_1^2 = \sigma_2^2$

then $F = \frac{s_1^2}{s_2^2} \sim F_{n-1, n-1}$

$\therefore$ F dist. is c/d variance ratio distribuⁿ

$$F_{\alpha}(v_1, v_2) = \frac{1}{F_{1-\alpha}(v_2, v_1)}$$

- Populaⁿ → parameters μ, σ^2, p
 Sample → statistics $\bar{X}, s^2, \hat{p}$
 populaⁿ → sample → statistics → inference

- $X_1, \dots, X_n \sim \text{iid}$ then $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$
 $E[X_i] = \mu$
 $\text{Var}(X_i) = \sigma^2$
 $E[\bar{X}] = \mu$
 $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$
 $SD = SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$

- if populaⁿ is normal $X_i \sim N(\mu, \sigma^2)$ then
 $\bar{X} \sim N(\mu, \sigma^2/n)$ $\therefore, \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right) \sim N(0, 1)$

CLT for large n , even if populaⁿ was not normal - then $\bar{X} \approx N(\mu, \sigma^2/n)$ $\therefore, \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right) \approx N(0, 1)$
 $n > 30$

- $SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$ if σ is not known $\left[SE(\bar{X}) = \frac{s}{\sqrt{n}} \right]$

- populaⁿ proportion is p , then $\hat{p} = \frac{X}{n}$ $\rightarrow$ no. of success
 $E[\hat{p}] = p$, $\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$

- mean, σ known $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ $\xrightarrow{\text{statistic}}$ $\xrightarrow{\text{dist.}}$ $N(0, 1)$

- mean, σ unknown
 normal populaⁿ $\frac{\bar{X} - \mu}{s/\sqrt{n}}$ $\xrightarrow{\text{statistic}}$ t_{n-1}

- 1 variance $\frac{(n-1)s^2}{\sigma^2}$ $\xrightarrow{\text{statistic}}$ χ^2_{n-1}

- 2 var. $\frac{s_1^2/\sigma_1^2}{s_2^2/\sigma_2^2}$ $\xrightarrow{\text{statistic}}$ F_{n_1-1, n_2-1}

- proportion $\hat{p}$ $\xrightarrow{\text{statistic}}$ normal

- Estimator is a statistic used to estimate θ
 $\hookrightarrow \hat{\theta} \quad \theta = \mu / \sigma^2 / \rho$
 $\hat{\theta} = \bar{X} / S^2 / \hat{\rho}$
- Point estimator gives 1 value. Interval estimator gives a range of values.
- An estimator $\hat{\theta}$ is unbiased for θ if $E[\hat{\theta}] = \theta$

Eq. Is $\bar{X}$ unbiased for μ ?
 Yes, why $E[\bar{X}] = \mu$

Eq. Is S^2 unbiased for σ^2 ?

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \quad \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

$$E[S^2] = \sigma^2, \quad \text{Yes}$$

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta \quad \text{if } E[\hat{\theta}] = \theta \text{ then bias is 0.}$$

$$E[\hat{\sigma}^2] = \frac{n-1}{n} \sigma^2, \quad \text{bias}(\hat{\sigma}^2) = \frac{n-1}{n} \sigma^2 - \sigma^2 = -\frac{\sigma^2}{n}$$

$$\text{Bias}(\hat{\sigma}^2) = -\frac{\sigma^2}{n}$$

$$\text{if } |\text{Bias}(\hat{\theta}_n)| \rightarrow 0, \text{ as } n \rightarrow \infty \text{ then}$$

estimator is c/d asymptotically unbiased